PROJECT COMPETITION 2026

Department of Information Technology | Techno Main Salt Lake

- **Problem Statement Title –** Develop a system to securely encrypt and decrypt files cryptographic algorithms.
- **Team Number -** 41
- **Team Name –** Secure Minds
- **Student Name – University Roll with Year of Registration**
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Secure File Encryption and Decryption System (CRYPTEX VAULT)

The Problem:

- ❑ Rising risk of data theft and cyber attacks
- ❑ Sensitive files can be accessed without authorization
- ❑ Existing storage systems often lack strong encryption
- ❑ Users require a secure and easy-to-use protection system

Proposed Solution:

CRYPTEX VAULT

- ❑ Intelligent File Security Platform
- ❑ A web-based application that enables users to securely encrypt and decrypt files using modern cryptographic techniques.

Core Features:

- ❑ AES-based symmetric file encrypt decryption
- ❑ Secure password-based authentic
- ❑ Strong confidentiality and data pr
- ❑ Unauthorized access prevention
- ❑ User-friendly interactive interface
- ❑ Secure cryptographic key manage
- ❑ Efficient and scalable security arc

TECHNICAL APPROACH

Tech stack:

- **Frontend:** HTML + CSS
- **Backend**: Python + Flask
- **Security:** Password-based key derivation using PBKDF2+Salt
- **Encryption** : Fernet (AES-based encryption)
- **Deployment:** Render (for hosting the web application)
- **Version Control**: Git + GitHub

❖ Implementation

FEASIBILITY AND VIABILITY

Technical Feasibility :

- ❑ AES-based encryption implemented using Fernet library
- ❑ Flask enables lightweight and efficient web deployment
- ❑ Supports multiple file formats for encryption/decryption
- ❑ Random salt generation to prevent identical passwords from generating identical encryption keys.

Economic Feasibility :

- ❑ Low development and maintenance cost
- ❑ Open-source technologies reduce expenses
- ❑ Zero licensing cost

Operational Feasibility:

- ❑ Easy for users to operate
- ❑ Minimal training required
- ❑ Simple and user-friendly interface
- ❑ Minimal technical knowledge required for users

Security Feasibility :

- ❑ Password-protected file encrypti
- ❑ No permanent storage of files or

Future Viability :

- ❑ Can be extended with cloud stor authentication
- ❑ Scalable for advanced cybersec

IMPACT AND BENEFITS

Cybersecurity Benefits:

- AES-based encryption protects confidential files from unauthorized access
- Password-protected decryption ensures secure file recovery
- Prevents data leakage during online file sharing

User Benefits:

- Simple and easy-to-use web interface
- Fast file encryption and decryption process
- Supports files like PDF, DOCX, ZIP, images, and text files

Technical Benefits:

- Built using Flask and Fernet (AES-basec
- In-memory processing avoids permane
- Lightweight architecture suitable for cl

Business & Practical Benefits:

- Low development and deployment co technologies
- Useful for students, organizations, anc security
- Scalable for future cybersecurity and c applications

RESEARCH AND REFERENCES

Cryptography Research :

- ❑ Studied AES-based encryption and Fernet cryptographic implementation
- ❑ Analyzed password-based encryption and SHA-256 hashing techniques
- ❑ Explored secure file protection methods against unauthorized access

Libraries/Docs :

- ❑Flask Documentation — https://flask.palletsprojects.com/
- ❑Cryptography Library Docs — https://cryptography.io/
- ❑Python Documentation — https://docs.python.org/3/
- ❑Render Deployment Docs — https://render.com/docs

References:

- ❑ Python Cryptography Official Do
- ❑ NIST (National Institute of Technology) AES Standards

Project Deployment:

- ❑ Successfully deployed on **“R**
- ❑ Publicly accessible web appli
- ❑ Supports real-time file encryp

Website Link:
https://fileencryption-crypto.onre